Chapter 8

8. The general linear model.

8.1. Definition and properties.

8.2. Linear hypothesis.

8.3. MLE.

8.4. A projection approach.

8.5. Distribution theory.

8.6. Inference.

8.7. Residual analysis.

8.7. Real data example.

The linear hypothesis

Definition

Let Y be an n-r.v. with $EY = \mu$. A linear model of dimension k is a hypothesis for μ of the form $H_0 : \mu \in L$, where L is a linear subspace of $\mathbb{R}^n$ with with $dim(L) = k$.

The linear hypothesis

For a given basis

$$X = (x_1, \ldots, x_k) = \begin{pmatrix} x_{11} & \cdots & x_{1k} \\ x_{21} & \cdots & x_{2k} \\ \vdots & \cdots & \vdots \\ x_{n1} & \cdots & x_{nk} \end{pmatrix}$$

of L this means that under H_0 there is a coefficient vector $\beta = (\beta_1, \ldots, \beta_k)'$ such that

$$\mu = \sum_{i=1}^{k} \beta_i x_i = X\beta.$$

The linear hypothesis

Writing

$$Y = \mu + \underbrace{(Y - \mu)}_{\varepsilon}$$

we have

$$Y = X\beta + \varepsilon,$$

where $E\varepsilon = 0$ (and quite often it is assumed $\mathrm{Var}(\varepsilon) = \sigma^2 I_n$).

The linear hypothesis

There exists also a multivariate version:

If Y_i' are p-r.v. and $Y = (Y_1', \ldots, Y_n')$ we have a hypothesis of the from

$$EY = M = XB,$$

where $M \in \mathbb{R}^{n \times p}$, $X \in \mathbb{R}^{n \times k}$ and $B \in \mathbb{R}^{k \times p}$.

Here we only discuss the case where the responses Y_i are scalar.

The linear hypothesis

Examples.

- Y_i are i.i.d. with mean β_0.

$$\mu = \begin{pmatrix} \beta_0 \\ \beta_0 \\ \vdots \\ \beta_0 \end{pmatrix} = \underbrace{\begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix}}_{X} \underbrace{\beta_0}_{\beta}.$$

Estimation and inference reduces to well known procedures.

The linear hypothesis

Examples.

- $Y = (V_1', \ldots, V_m')'$, where $V_i = (Y_{i1}, \ldots, Y_{ig_i})'$ with $EY_{ij} = \beta_i$ and $n = g_1 + \cdots + g_m$. Then

$$\mu = \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_2 \\ \vdots \\ \beta_m \\ \vdots \\ \beta_m \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 0 & 0 & \ldots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & 0 & \ldots & 0 \\ 0 & 1 & 0 & \ldots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 1 & 0 & \ldots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \ldots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \ldots & 1 \end{pmatrix}}_{X} \underbrace{\begin{pmatrix} \beta_1 \\ \vdots \\ \beta_m \end{pmatrix}}_{\beta}.$$

The linear hypothesis

Examples, continued.

- The mean of Y_i depends on several (known) variables (*dependent variables*, *regressors*) $x_1, \ldots, x_k$ in a linear way:

$$\mu = \underbrace{\begin{pmatrix} x_{11} & \ldots & x_{1k} \\ x_{21} & \ldots & x_{2k} \\ \vdots & \ldots & \vdots \\ x_{n1} & \ldots & x_{nk} \end{pmatrix}}_{X} \underbrace{\begin{pmatrix} \beta_1 \\ \vdots \\ \beta_m \end{pmatrix}}_{\beta}$$

The linear hypothesis

Main targets.

- Estimate the unknown parameters β (or μ respectively) and σ^2 if $\mathrm{Var}(\varepsilon) = \sigma^2 I_n$.
- Establish the distribution of the estimators when Y is multivariate normal vector.
- Test different hypotheses for β (or μ respectively). E.g. $\beta_\ell = 0$ or $\beta_1 = \beta_2 = \cdots = \beta_k$ or $\mu \in L_2 \subset L_1$ ...

MLE

Let us assume $Y \sim \mathcal{N}_n(\mu, \sigma^2 I_n)$. Then the likelihood function is

$$\ell(\mu, \sigma | Y) = \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right)^n \exp\left(-\frac{1}{2\sigma^2} \| Y - \mu \|^2 \right).$$

If the linear model $\mu = X\beta$ holds, then then the MLE for (β, σ^2) is

$$(\hat{\beta}, \hat{\sigma}^2) = \operatorname{argmax}_{\beta, \sigma^2} \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right)^n \exp\left(-\frac{1}{2\sigma^2} \| Y - X\beta \|^2 \right).$$

MLE

Theorem

Suppose we have $Y \sim \mathcal{N}_n(X\beta, \sigma^2 I_n)$. *Then the MLEs for* β *and* σ^2 *are given by*

$$\hat{\beta} = (X'X)^{-1}X'Y \quad \textit{and} \quad \hat{\sigma}^2 = \frac{1}{n}\|Y - X\hat{\beta}\|^2.$$

The maximum of the likelihood is

$$\left(\frac{1}{2\pi e}\right)^{n/2} \left(\hat{\sigma}^2\right)^{-n/2}.$$

□

MLE

Remark. The estimators

$$\hat{\beta} = (X'X)^{-1}X'Y \quad \text{and} \quad \hat{\sigma}^2 = \frac{1}{n}\|Y - X\hat{\beta}\|^2.$$

also make sense, when Y is not normally distributed. In this case $\hat{\beta}$ above, is the so-called *least-squares estimator.*

The vector $\hat{\varepsilon} := Y - \hat{\beta}X$ is called the *residual vector*. We will write

$$SSR = \|Y - X\hat{\beta}\|^2.$$

(SSR stands for sum of squared residuals).

A projection approach

A way to obtain the MLE for β is to write

$$\hat{\mu} = X\hat{\beta} = \text{argmin}_{\mu \in L} \| Y - \mu \|,$$

where L is the linear space spanned by vectors $(x_1, \ldots, x_k)$. It follows that $\hat{\mu} = p_L(Y)$ the *orthogonal projection* of Y onto L.

Note that the orthogonal projection is characterized as the unique element $\hat{\mu} \in L$ such that

$$x'(Y - \hat{\mu}) = 0 \quad \text{for all} \quad x \in L$$

$$\Longleftrightarrow$$

$$x_j'(Y - \hat{\mu}) = 0 \quad \text{for all} \quad j \in \{1, \ldots, k\}$$

$$\Longleftrightarrow$$

$$X'(Y - \hat{\mu}) = 0.$$

A projection approach

$$\Longleftrightarrow \qquad X'(Y - X\hat{\beta}) = 0.$$

$$\Longleftrightarrow \qquad X'Y = X'X\hat{\beta}$$

$$\Longleftrightarrow \qquad (X'X)^{-1}X'Y = \hat{\beta}.$$

Hence we have

$$\hat{\mu} = p_L(Y) = X(X'X)^{-1}X'Y = X\hat{\beta}.$$

A projection approach

The matrix

$$H = X(X'X)^{-1}X'$$

is called the *projection matrix* or *hat matrix*. It is (exercise)

- symmetric: $H = H'$, and
- idempotent: $H^2 = H$.

Distribution theory

The proposition below holds without assuming normality.

Proposition

Assume that Y is a linear model of dimension k of the from

$$Y = \mu + \varepsilon = X\beta + \varepsilon \quad \varepsilon \sim (0, \sigma^2 I_n).$$

Let $\hat{\mu}$, $\hat{\beta}$ and $\hat{\sigma}^2$ be defined as before. Then

$$E\hat{\beta} = \beta \quad \textit{and} \quad E\hat{\mu} = \mu;$$
$$\mathrm{Var}(\hat{\beta}) = \sigma^2(X'X)^{-1} \quad \textit{and} \quad \mathrm{Var}(\hat{\mu}) = \sigma^2 H;$$
$$E\hat{\sigma}^2 = \frac{n-k}{n}\sigma^2.$$

Distribution theory

Proposition

Assume that Y is a linear model of dimension k of the from

$$Y = \mu + \varepsilon = X\beta + \varepsilon \quad \varepsilon \sim \mathcal{N}_n(0, \sigma^2 I_n).$$

Let $\hat{\mu}$, $\hat{\beta}$ and $\hat{\sigma}^2$ be defined as before. Then

(*i*) $\hat{\beta} \sim \mathcal{N}_k(\beta, \sigma^2(X'X)^{-1})$;

(*ii*) $\hat{\mu} \sim \mathcal{N}_n(\mu, \sigma^2 H)$;

(*iii*) $\hat{\sigma}^2 \sim \frac{\sigma^2}{n}\chi^2_{n-k}$;

(*iv*) $\hat{\sigma}^2$ *and* $\hat{\beta}$ *(or* $\hat{\mu}$*, respectively) are independent.*

Distribution theory

The above result basically follows from the following:

Proposition

Let $L_1, L_2, \ldots, L_r$ be orthogonal subspaces of $\mathbb{R}^n$ with $\dim(L_i) = k_i$ and $L_1 \oplus L_2 \oplus \cdots \oplus L_r = \mathbb{R}^n$ and let $\varepsilon \sim \mathcal{N}_n(0, \sigma^2 I_n)$. Then

$$(a) \quad p_{L_1}(\varepsilon), \ldots, p_{L_r}(\varepsilon) \quad \textit{are independent.}$$

$$(b) \quad \|p_{L_i}(\varepsilon)\|^2 \sim \sigma^2 \chi^2_{k_i} \quad 1 \leq i \leq r.$$

Inference

For a linear model $Y \sim \mathcal{N}_n(\mu, \sigma^2 I_n)$ we will test hypotheses of the following form:

$$H_0 : \mu = \sum_{i=1}^{k} x_i \beta_i,$$

against

$$H_A : \mu = \sum_{i=1}^{\ell} x_i \beta_i, \quad k < \ell \leq n.$$

Or more generally

$$H_0 : \mu \in L_2 \quad \text{against} \quad H_A : \mu \in L_1 \backslash L_2 \quad (L_2 \subset L_1).$$

We assume in the sequel that $\dim(L_1) = k_1$ and $\dim(L_2) = k_2 \leq k_1$.

Inference

The likelihood ratio test is

$$\Lambda(Y) = \frac{\sup_{\mu_2 \in L_2, \sigma_2^2 > 0} \ell(\mu_2, \sigma_2^2 | Y)}{\sup_{\mu_1 \in L_1, \sigma_1^2 > 0} \ell(\mu_1, \sigma_1^2 | Y)}.$$

Recall that the maximum of the likelihood for the model L is proportional to

$$(\hat{\sigma}^2)^{-n/2} = \left(\frac{1}{n} \| Y - \hat{\mu} \|^2 \right)^{-n/2}.$$

Inference

The LR test statistic is

$$\Lambda(Y) = \left(\frac{\|Y - \hat{\mu}_1\|^2}{\|Y - \hat{\mu}_2\|^2} \right)^{n/2},$$

and rejects if the $\Lambda(Y)$ is too small, or equlvalently if

$$\frac{\|Y - \hat{\mu}_2\|^2}{\|Y - \hat{\mu}_1\|^2}$$

is too large.

Inference

Now since

$$\|Y - \hat{\mu}_2\|^2 = \|Y - \hat{\mu}_1 + \underbrace{\hat{\mu}_1 - \hat{\mu}_2}_{\in L_1}\|^2$$

we get by the projection theorem and Pythagoras' theorem that

$$\|Y - \hat{\mu}_2\|^2 = \|Y - \hat{\mu}_1\|^2 + \|\hat{\mu}_1 - \hat{\mu}_2\|^2.$$

Hence the LR test is equivalent to the test which rejects if

$$F(Y) = \frac{\|\hat{\mu}_1 - \hat{\mu}_2\|^2/(k_1 - k_2)}{\|Y - \hat{\mu}_1\|^2/(n - k_1)}$$

is too big.

Inference

Theorem

Assume a linear model of the form $Y \sim \mathcal{N}_n(\mu, \sigma^2 I_n)$. *The likelihood ratio test for* $H_0 : \mu \in L_2$ *against* $H_A : \mu \in L_1/L_2$ *is equivalent to an* F *test. Under* H_0 *the test statistic*

$$F(Y) = \frac{\|\hat{\mu}_1 - \hat{\mu}_2\|^2/(k_1 - k_2)}{\|Y - \hat{\mu}_1\|^2/(n - k_1)}$$

follows an F*-distribution with* $k_1 - k_2$ *and* $n - k_1$ *degrees of freedom. At significance level* α *we reject* H_0 *if*

$$F(Y) \geq F_{k_1-k_2, n-k_1; 1-\alpha}.$$

Inference

Remark. Let us define

$$\tilde{\sigma}_1^2 = \frac{1}{n - k_1} \| Y - \hat{\mu}_1 \|^2.$$

Note that this is an unbiased estimator for σ^2 for the model L_1, and that under H_0 as well as under H_1 this converges to σ^2.

Then we can write

$$F(Y) = \frac{1}{k_1 - k_2} \frac{\| \hat{\mu}_1 - \hat{\mu}_2 \|^2}{\tilde{\sigma}_1^2}.$$

Analysis of the residuals

A common approach to check if an appropriate model has been fit the data is to investigate the residuals.

If y_k are the observed variables and $\hat{y}_k$ are the fitted values from a linear model, then

$$\hat{\varepsilon}_k := y_k - \hat{y}_k$$

are called residuals.

For example, in a linear regression the fitted values are $\hat{y}_k = x_k\hat{\beta}$. So the residuals can be seen as estimates of the model errors ε_k. Accordingly, if the model has been correctly specified, the distribution of the $\hat{\varepsilon}_k$'s is expected to resemble the distribution imposed on the ε_k's. Let us verify that this is true.

Analysis of the residuals

Suppose we have the linear model $Y \sim \mathcal{N}_n(\mu, \sigma^2 I_n)$, with $\mu \in L$ under H_0. The residual vector is given by

$$R = Y - \hat{\mu} = Y - HY = (I_n - H)Y = (I_n - H)(\mu + \varepsilon) \overset{H_0}{=} (I_n - H)\varepsilon.$$

Hence

$$ER = 0 \quad \text{and} \quad \mathrm{Var}(R) = \sigma^2(I_n - H),$$

Showing that

$$R \sim \mathcal{N}_n(0, \sigma^2(I_n - H)).$$

Note also that $\mathrm{Cov}(\hat{\mu}, R) = 0$ and hence $\hat{\mu}$ and R are independent.

Analysis of the residuals

As we have mentioned above, R is in some sense an estimator for the non-observed errors-vector ε.

We have

$$\varepsilon - R = H\varepsilon,$$

and

$$\mathrm{Var}(\varepsilon - R) = \sigma^2 H.$$

We see that

$$\frac{1}{n}\sum_{i=1}^{n} E(\varepsilon_i - R_i)^2 = \frac{1}{n}\sum_{i=1}^{n} \mathrm{Var}(\varepsilon_i - R_i) = \frac{\sigma^2}{n}\mathrm{tr}(H) = \frac{k\sigma^2}{n},$$

so that in average the mean square distance between ε_i and R_i is of order $1/n$.

Analysis of the residuals

Example. Let us consider a simple linear regression with

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad \varepsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2),$$

where $1 \leq i \leq n$. We can rewrite this as

$$Y_i = \alpha_0 + \beta_1 t_i + \varepsilon_i,$$

where $\alpha_0 = \beta_0 + \beta_1 \bar{x}$ and $t_i = x_i - \bar{x}$. Hence $\bar{t} = 0$. Let us denote $t_+ = \sum t_i^2$.

In vector form our model corresponds to

$$Y = X\beta + \varepsilon, \quad \text{with} \quad X = (1_n, t) \quad \text{and} \quad \beta = (\alpha_0, \beta_1),$$

where $1_n = (1, \ldots, 1)' \in \mathbb{R}^n$. Hence $1_n' t = n\bar{t} = 0$.

Analysis of the residuals

In this example $X'X = \begin{pmatrix} n & 0 \\ 0 & t_+ \end{pmatrix}$. This shows that $\hat{\alpha}_0$ and $\hat{\beta}_1$ are independent. Moreover, we have

$$H = X(X'X)^{-1}X' = \left(\left(\frac{1}{n} + \frac{t_i t_j}{t_+} \right) \right)_{i,j=1}^{n}.$$

If we assume that the x_i are realizations of some i.i.d. sequence with variance σ_x^2, then by the law of large numbers $t_+ \sim n\sigma_x^2$ and hence $H_{ij} = O_P\left(n^{-1}\right)$ for all $1 \leq i \leq j \leq n$.

In particular, the elements of H in the diagonal, $\varepsilon_i - R_i$, are close to zero if n is big.

Analysis of the residuals

Moreover,

$$\mathrm{Var}(R) = \sigma^2(I_n - H),$$

we see that

$$\mathrm{cov}(R_i, R_j) \approx \begin{cases} \sigma^2 & \text{if } i = j \\ 0 & \text{else} \end{cases}$$

if n is large. This is in line with $\mathrm{Var}(\varepsilon) = \sigma^2 I_n$.

Real data example

For selling firewood it is important to know the volume of a tree, which is not so easy to measure. The simplest method is to measure diameter and hight and base an estimate on the volume on these parameters.

The numbers below show a sample of 31 Pennsylvanian cherry trees (data set `trees` in R). (Girth in inches, hight in feed and volume in cubic feet).

```
> head(trees)
  Girth Height Volume
1   8.3     70   10.3
2   8.6     65   10.3
3   8.8     63   10.2
4  10.5     72   16.4
5  10.7     81   18.8
6  10.8     83   19.7
...
```

Real data example

We first consider a linear model of the form

$$Y = X\beta + \varepsilon, \quad \varepsilon \sim \mathcal{N}_n(0, \sigma^2 I_n),$$

where

$$X = \begin{pmatrix} 1 & 8.3 & 70 \\ 1 & 8.6 & 65 \\ 1 & 8.8 & 63 \\ \vdots & \vdots & \vdots \end{pmatrix} \in \mathbb{R}^{31\times 3}.$$

The 1's in the first column give a model with intercept.

In `R` we can use the package `lm` to estimate the model.

Real data example

```
> lin.mod <- lm(Volume~1+Girth+Height,data=trees)
> summary(lin.mod)

Call:
lm(formula = Volume ~ 1 + Girth + Height, data = trees)

Residuals:
    Min      1Q  Median      3Q     Max
-6.4065 -2.6493 -0.2876  2.2003  8.4847

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -57.9877     8.6382  -6.713 2.75e-07 ***
Girth         4.7082     0.2643  17.816  < 2e-16 ***
Height        0.3393     0.1302   2.607   0.0145 *
---
Signif. codes:  0 *** 0.001 ** 0.01 * 0.05 . 0.1  1

Residual standard error: 3.882 on 28 degrees of freedom
Multiple R-squared:  0.948,Adjusted R-squared:  0.9442
F-statistic:   255 on 2 and 28 DF,  p-value: < 2.2e-16
```

Real data example

What do we get from the summary?

Residuals: We obtain the emprirical *quartiles* of the estimated residuals and the two extreme values.

Coefficients: We have four columns. `Estimate` corresponds to $\hat{\beta}$, `Std. Error` is the estimated standard deviation of $\hat{\beta}$, i.e. $\tilde{\sigma} \times \mathrm{diag}(X'X)^{-1/2}$, with $\tilde{\sigma}^2 = MSR = \frac{1}{n-k} SSR$, in our case $k = 3$. Then, a t-test is used in order to determine the influence of a given variable. The absolute value of the t-value is $\sqrt{F(y)}$, where

$$F(y) = \|\hat{\mu}_1 - \hat{\mu}_2\|^2 / MSR$$

and $\hat{\mu}_1$ is the estimated mean under the complete model, while $\hat{\mu}_2$ is the estimated mean under the linear hypothesis which excludes one parameter.

Real data example

`Pr(>|t|)` is the p-valeur, i.e. the probability that under H_0 the test statistic exceeds its realization. One can show that the t-test and the F-test are equivalent, and hence $P(F(Y) > F(y))$.

Residual standard error: corresponds to $\sqrt{MSR} = \sqrt{SSR/28}$.

Multiple R-squared: goodness-of-fit measure. It corresponds to

$$R^2 = 1 - \frac{SSR}{SST},$$

where $SST = \sum_{i=1}^{n}(Y_i - \bar{Y})^2$ is *the total sum of squares*.

Real data example

Adjusted R-squared: a measures for *goodness-of-fit* which penalizes for *over-parametrisation*. It equals

$$\bar{R}^2 = 1 - \frac{SSR(n-1)}{SST(n-k-1)},$$

where k is the dimension of the model.

F-statistic: Value of the F-statistic for the test $H_0 : \mu \in L_2$ against $H_1 : L_1$, where

$$L_2 = \mathrm{sp}\{1_n\} \quad \text{et} \quad L_1 = \mathrm{sp}\{X\}.$$

Real data example

According to the analysis all the included parameters are strongly significant, and the model gives an excellent fit.

Let us check if the model assumption do hold and analyse the residuals.

Real data example

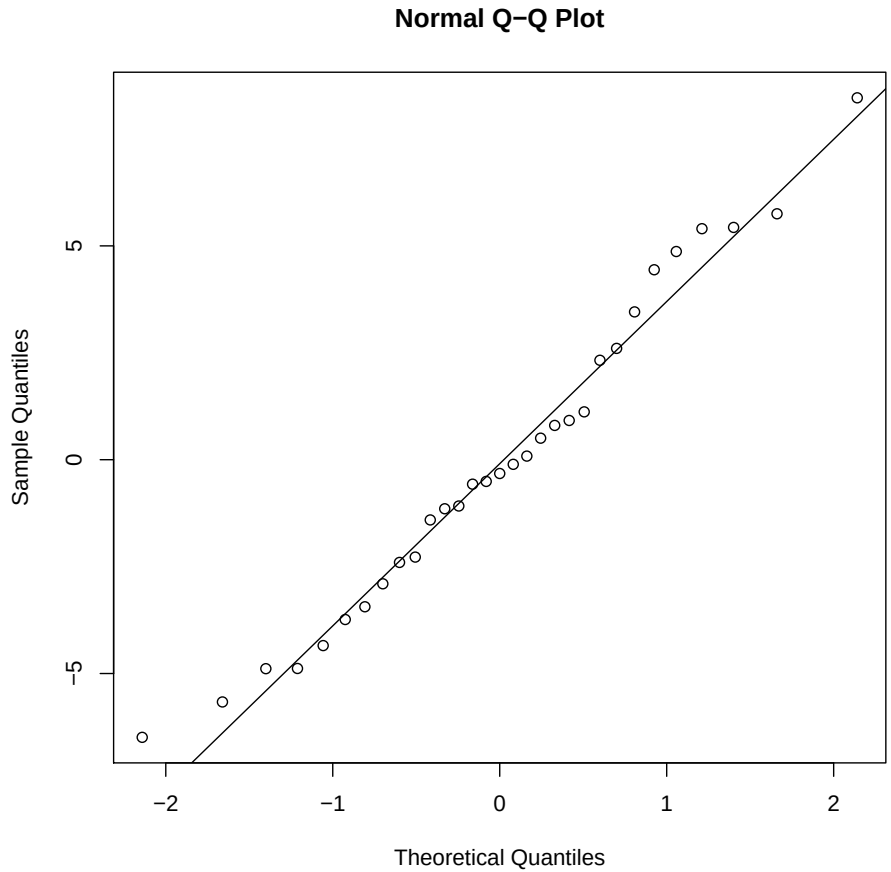


Figure 1: The qqplot against normal quantiles shows no serious departure from normal.

Real data example

We observe a "V"-shape in the plots of fitted values and diameter against residuals.

This indicates that a linear model is not appropriate!

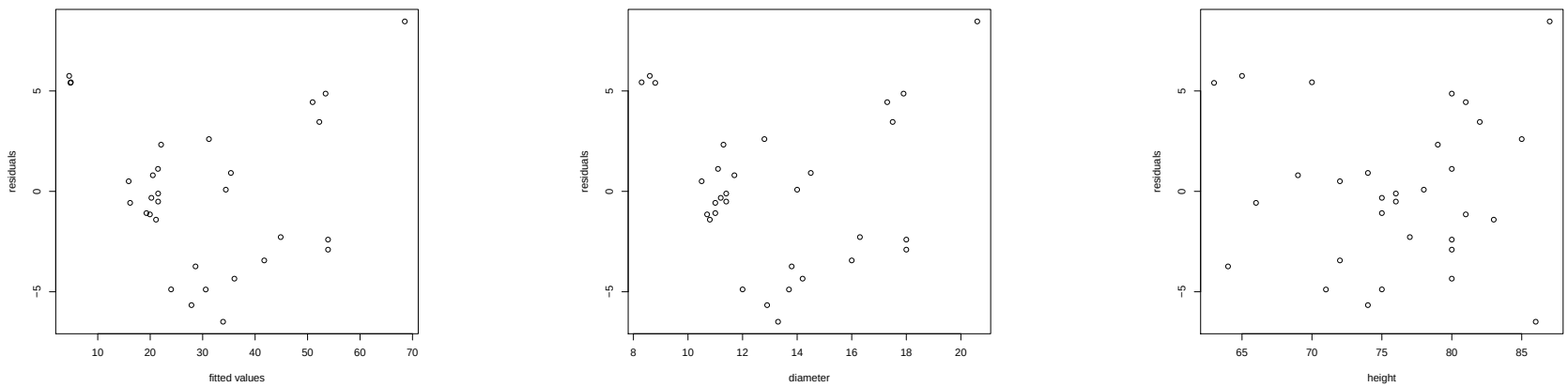


Figure 2: We plot fitted values (left), diameter (middle) and height (right) against residuals.

Real data example

It seems reasonable to impose a cylinder or cone as the rough shape of a tree.

$$v = \frac{\pi d^2 h}{4} \quad \text{or} \quad v = \frac{\pi d^2 h}{12}.$$

So a log-transformation could be reasonable:

$$\log V = \beta_0 + \beta_1 \log d + \beta_2 \log h + \varepsilon.$$

Real data example

We convert inches to feet for the diameter and take logarithms.

```
> log.vol <- log(trees$Volume)
> log.height <- log(trees$Height)
> log.girth <- log(trees$Girth)
```

The new design matrix is

$$X = \begin{pmatrix} 1 & -1.56 & 3.06 \\ 1 & -1.52 & 2.99 \\ 1 & -1.50 & 2.96 \\ \vdots & \vdots & \vdots \end{pmatrix} \in \mathbb{R}^{31 \times 3}.$$

Real data example

```
> lin.mod2 <- lm(log.vol~log.girth+log.height,data=trees)
> summary(lin.mod2)

Call:
lm(formula = log.vol ~ log.girth + log.height, data = trees)

Residuals:
      Min        1Q    Median        3Q       Max
-0.168561 -0.048488  0.002431  0.063637  0.129223

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -6.63162    0.79979  -8.292 5.06e-09 ***
log.girth    1.98265    0.07501  26.432  < 2e-16 ***
log.height   1.11712    0.20444   5.464 7.81e-06 ***
---
Signif. codes:  0 *** 0.001 ** 0.01 * 0.05 . 0.1  1

Residual standard error: 0.08139 on 28 degrees of freedom
Multiple R-squared:  0.9777,Adjusted R-squared:  0.9761
F-statistic: 613.2 on 2 and 28 DF,  p-value: < 2.2e-16
```

Real data example

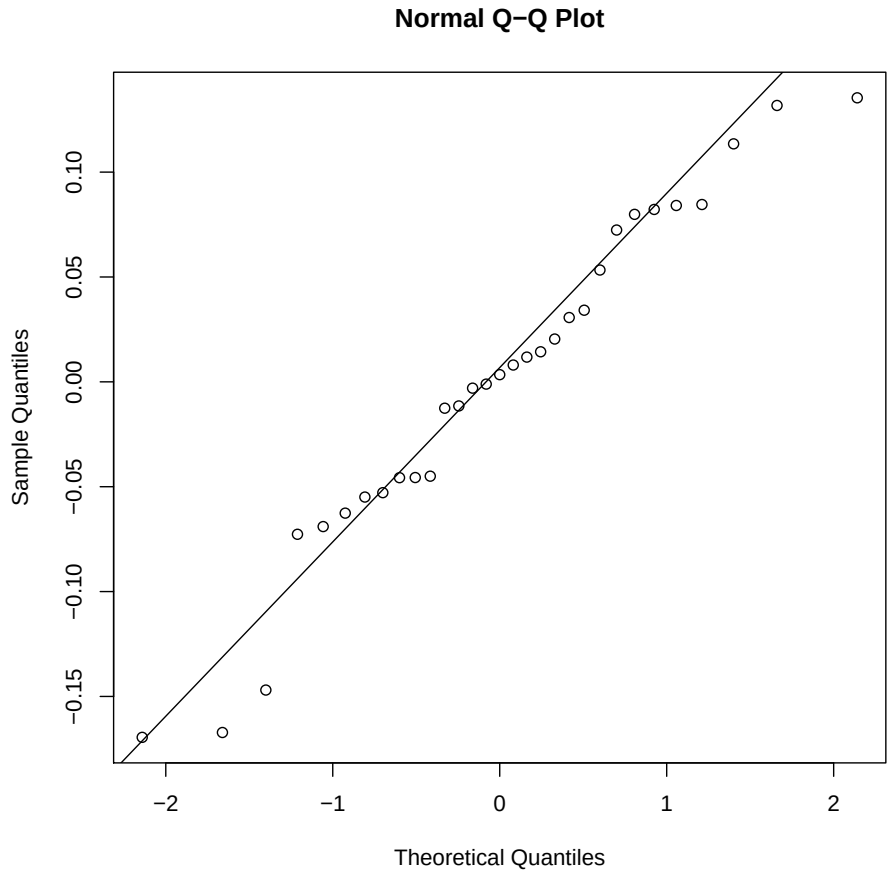


Figure 3: The qqplot against normal quantiles shows no serious departure from normal.

Real data example

No residual plot reveals obvious departures from the model assumption:

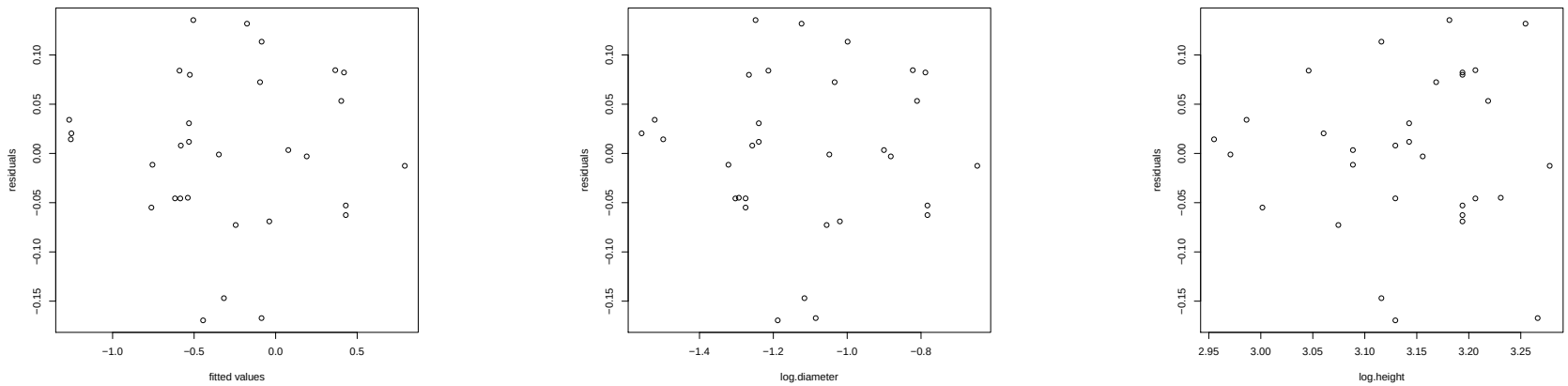


Figure 4: We plot fitted values (left), diameter (middle) and height (right) against residuals.

Remark: Let us impose a cone on the tree

$$v = \frac{\pi d^2 h}{12} \Longrightarrow \log v = -1.34 + 2 \log d + \log h,$$

and compare the estimated coefficients.